

First-Price Auction

FPA in Independent Private Value Environments

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This talk

Static Games with **In**complete Information

- type-dependent strategies
- Bayesian Nash Equilibrium

First-Price Auctions

- FPA in IPV Environment
- realized profit

Comparing Auctions

- Strategic Equivalence in IPV Environments
- Collusion

Static Games with Incomplete Information

Essentials

- static game: $N = \{1, 2, \dots, n\}$ and $\{S_1, S_2, \dots, S_n\}$
- incomplete information (about payoffs): (ex post) payoff of a player i is now a *random* variable, and Nature gives the player i information about her payoff (and possibly about payoffs of the opponents), summarized in her *type* θ_i
- type-space of player i : $\theta_i \in \Theta_i = \{\theta_{i1}, \theta_{i2}, \dots, \theta_{ik_i}\}$
- i.e., a *finite* number k_i of types of player i
- type-dependent payoff of player from a profile of strategies s is $v_i(s; \theta_i)$
- $\phi(\theta_{-i}|\theta_i)$ is the posterior/conditional probability of θ_{-i} (all other types but i)
- ! sometimes we study a case where each type θ_i reveals to player i only information about her payoff and not about the opponents' payoffs (all posterior distributions of θ_{-i} are equal to the prior distribution); sometimes not
- both prior and posterior (unconditional and conditional) distributions of types are common knowledge

Essentials

Timeline

1. Nature chooses a profile of types $(\theta_1, \dots, \theta_n)$.
2. Each player i privately learns own type θ_i (and constructs her posterior over the opponents' types $\phi(\theta_{-i}|\theta_i)$).
3. Players simultaneously choose type-dependent pure strategies $s_i(\theta_i) \in S_i$.
4. Players receive the (interim expected) payoff $\mathbb{E}[v_i(s; \theta_i)]$.

Bayesian Nash Equilibrium (in pure strategies)

Definition

A strategy profile $(s_1^*, s_2^*, \dots, s_n^*)$ is a **pure-strategy Bayesian Nash equilibrium** if, for every player i and for each player i 's type $\theta_i \in \Theta_i$:

$$\sum_{\theta_{-i} \in \Theta_{-i}} \phi_i(\theta_{-i} | \theta_i) v_i(s_i^*(\theta_i), s_{-i}^*(\theta_{-i}); \theta_i) \geq \sum_{\theta_{-i} \in \Theta_{-i}} \phi_i(\theta_{-i} | \theta_i) v_i(s_i, s_{-i}^*(\theta_{-i}); \theta_i), s_i \in S_i$$

Bayesian Nash Equilibrium

- It is NE (mutual best responses) extended to incomplete information.
- $s_{-i}^*(\theta_{-i})$ is now a vector of the opponents' strategies, where each opponent selects her strategy depending on her privately observed type.
- Like a NE in pure strategies may not exist, also a BNE in pure strategies may not exist.
- We can extend to a continuum of types and a continuum of actions.
- Typically, payoffs are monotonic (e.g., increasing) in type, and therefore also the equilibrium strategy is monotonic (e.g., increasing) in type.

FPA in IPV Environment

Recall (Symmetric) IPV Environment

- 1 indivisible object
- N potential buyers without any liquidity constraints
- X_i is a buyer i 's stochastic valuation of the object (fundamental)
- $X_i \sim F$ are i.i.d. (independently and identically distributed)
- F is continuous, increasing and twice differentiable on an interval $[0, \omega]$
- each buyer i observes X_i but doesn't learn anything about $X_j, j \neq i$

FPA Mechanism

- Each player i submits a sealed bid $b_i \geq 0$.
- The highest bidder wins the object and pays the highest (own) price.
- For a profile $b = (b_1, \dots, b_N)$, the (win or loss) payoffs are

$$\Pi_i(x_i, b) = \begin{cases} x_i - b_i, & \text{if } b_i > \max_{j \neq i} b_j \\ 0, & \text{if } b_i < \max_{j \neq i} b_j. \end{cases}$$

- Tie ($b_i = \max_{j \neq i} b_j$): Equal probability of winning and paying among the highest bidders.

A Symmetric Separating BNE

Let $\beta_i(x)$ be the bidding strategy of buyer i .

- That is, $\beta_i(x_i)$ is the bid of the type x_i of the buyer i .

We will consider only the following bidding strategy profiles:

- *symmetric*, $\beta(x) := \beta_1(x) = \beta_2(x) = \dots = \beta_N(x)$
- *increasing*, $\beta(x) > \beta(x')$ if $x > x'$
 - consequently, a *separating* strategy (each type gives a unique bid; types don't *pool* with other types)
 - consequently, an inverse function $\beta^{-1}(b)$ exists (through this function, one can recover the player's type with certainty)
- *differentiable* (no 'jumps')
- *non-negative profits*, $\beta(x) \in [0, x]$

A Symmetric Separating BNE

- $Y_{-i} = \max_{j \neq i} X_j$ is the value/type of the strongest opponent of buyer i (out of $N - 1$ opponents).
- In IPV environment, $Y_{-i} \sim G = F^{N-1}$.
- Since bidding strategies are symmetric and increasing, buyer i cares only about the bid of the strongest opponent, because this bid is the *highest* out of bids of all $N - 1$ opponents:

$$\beta(Y_{-i}) = \beta(\max_{j \neq i} X_j) = \max_{j \neq i} \beta(X_j)$$

A Symmetric Separating BNE

- When the buyer i of type x_i bids b_i , what is her expected payoff?
- Either win or loss (due to continuity, tie is with probability zero):
- Win if $b_i > \beta(Y_{-i})$ or equivalently $Y_{-i} < \beta^{-1}(b_i)$. Win occurs with the probability $G(\beta^{-1}(b_i))$.
- Loss if $b_i < \beta(Y_{-i})$ or equivalently $Y_{-i} > \beta^{-1}(b_i)$. Loss occurs with the probability $1 - G(\beta^{-1}(b_i))$.
- The expected payoff of buyer i of type x_i who gives a bid b_i is:

$$G(\beta^{-1}(b_i))(x_i - b_i)$$

Notice: For convenience, we henceforth avoid index in the subscript.

A Symmetric Separating BNE

- In BNE, each type's bid is her best response; it maximizes the (interim) expected payoff given the strategies of the opponents.
- Maximizing the payoff with respect to b yields the F.O.C.:

$$\frac{g(\beta^{-1}(b))}{\beta'(\beta^{-1}(b))} (x - b) - G(\beta^{-1}(b)) = 0$$

- Enter that in a symmetric profile, $b = \beta(x)$, or equivalently, $\beta^{-1}(b) = x$:

$$\frac{g(x)}{\beta'(x)} [x - \beta(x)] - G(x) = 0$$

$$xg(x) = G(x)\beta'(x) + g(x)\beta(x)$$

A Symmetric Separating BNE

- To obtain $\beta(x)$, we first integrate RHS on the interval $y \in [0, x]$:
- Observe that:

$$\frac{d}{dx} (G(x)\beta(x)) = G(x)\beta'(x) + g(x)\beta(x)$$

- Therefore, integrated RHS is:

$$\int_0^x [G(y)\beta'(y) + g(y)\beta(y)] dy = G(x)\beta(x) - \underbrace{G(0)\beta(0)}_0$$

A Symmetric Separating BNE

- Second, we introduce the integral of the LHS on the interval $y \in [0, x]$:

$$\int_0^x yg(y)dy + k$$

- It is equal to the integral of the RHS on the interval $y \in [0, x]$:

$$\int_0^x yg(y)dy + k = G(x)\beta(x)$$

- Since $\beta(0) = 0$ (non-negative profits), we have $k = 0$, and we can write:

$$\beta(x) = \frac{1}{G(x)} \int_0^x yg(y)dy = \mathbb{E}[Y \mid Y < x]$$

Notice: Since we dropped index i from x_i and b_i , we also dropped $-i$ from Y_{-i} .

Equilibrium Bids in FPA in a Symmetric IPV Environment

- The auction is efficient; the buyer with the highest valuation wins the object with certainty.
- My optimal bid is the *average* second highest value of the opponents, conditional on me having the highest value.

$$\beta(x) = \mathbb{E}[Y \mid Y < x]$$

- Intuitively, if the values suddenly get revealed, then the strongest opponent would be, *on average*, exactly discouraged from competing.
- This property echoes the environment with *known* asymmetric values. By bidding $b_1 = v_2$, stronger Bidder 1 exactly discourages weaker Bidder 2 (her strongest opponent).

Realized profit (Shading)

Each buyer solves a tradeoff.

- A higher bid is increasing win probability, $G(\beta^{-1}(b))$.
- + marginal gain
- A higher bid is decreasing the profit, $x - b$, realized in any win event.
- inframarginal losses

How is this tradeoff solved in the equilibrium?

- Specifically, what is the *realized* profit, $x - \beta(x)$?
- The realized (not expected!) profit is called 'shading'.

Realized profit (Shading)

- Using integration by parts (per partes)

$$\beta(x) = \frac{1}{G(x)} \int_0^x yg(y)dy = \frac{1}{G(x)} \left[[yG(y)]_0^x - \int_0^x G(y)dy \right]$$

$$\beta(x) = x - \int_0^x \frac{G(y)}{G(x)} dy = x - \int_0^x \left[\frac{F(y)}{F(x)} \right]^{N-1} dy$$

Notice: $\frac{G(y)}{G(x)} < 1$ for $y \in [0, x)$ and therefore $\int_0^x \frac{G(y)}{G(x)} dy < x$.

Shading: Uniform, N bidders

$F(x) = x$, N bidders, $x \in [0, 1]$

$$\beta(x) = x - \int_0^x \frac{y^{N-1}}{x^{N-1}} dy = x - \frac{1}{x^{N-1}} \left[\frac{y^N}{N} \right]_0^x = x - \frac{x}{N} = \frac{N-1}{N} x$$

- The realized profit (shading) is $\frac{x}{N}$.
- With higher competition (larger N), shading is reduced.

Shading: Exponential, 2 bidders

$$F(x) = 1 - e^{-2x}, 2 \text{ bidders}, x \in [0, \infty)$$

$$\begin{aligned}\beta(x) &= x - \frac{1}{1 - e^{-2x}} \int_0^x 1 - e^{-2y} dy = x - \frac{1}{1 - e^{-2x}} \left[x + \frac{1}{2} [e^{-2y}]_0^x \right] = \\ &= x - \frac{1}{1 - e^{-2x}} \left[x + \frac{1}{2} (e^{-2x} - 1) \right] = x - \frac{1}{2} \frac{2x - (1 - e^{-2x})}{1 - e^{-2x}} = \frac{1}{2} - \frac{xe^{-2x}}{1 - e^{-2x}}\end{aligned}$$

- $\beta(x) < \frac{1}{2} = \mathbb{E}[x]$.
- Shading of high-value bidders is **massive**!

FPA Fuckups

FPA Fuckup: Australia 1993

2 licences for TV

- no deposit (**zero** default cost)
- in the event of default: if the highest bid is not paid, it goes to the next-highest bid, etc.

What could go wrong?

- Winners were undercapitalized (dark-horse) bidders.
- They submitted multiple bids at 5-million intervals to guarantee win.
- Effectively, those who understand this strategy turned it into a second-price auction for themselves, while exploiting that the others are using their first-price strategy (i.e., shading).

UCom: 212 mil

- sequential defaults
- bought at 117 mil, then resale with profit 21 mil

Hi Vision: 177 mil

- sequential defaults, then dropped out
- other firms also defaulted, bought at 77 mil

Strategic Equivalence

Strategic Equivalence of Auction Formats Under IPV

Dutch auction and FPA

- In Dutch auction, all relevant bidding strategies are *limit* strategies (i.e., your strategy can be described as the *initial* price at which you buy).
- Why? No information is revealed during bidding, hence no need to update your limit during bidding.
 - Participation decision brings no information: (i) Ending participation by buying the good ends the auction. (ii) Dropping auction early by some types cannot be used as a misleading signal (i.e., certain types pretend to have a different value, and subsequently re-enter). Intuitively, as the price goes down in favor of bidders, none participating type is dropping early.
 - Outcry is not allowed so it cannot be used as a signal. (In Dutch auction, the price is decreased gradually = a clock auction).
- The player with the highest limit wins the object and pays her limit price.
- *limit* in Dutch auction = *bid* in FPA

Strategic Equivalence of Auction Formats Under IPV

English auction and SPA

- In English auction, not all relevant strategies are (initial) *limit* strategies.
 - First, if you can 're-enter' (open-exit auction), you can pretend to have a low value by dropping early (low-value types drop early because the price is increasingly less favorable). Later, you can re-enter.
 - Second, information about (low) values is revealed during bidding, and therefore the bidder may be tempted to adjust the initial limit.
 - Third, English auction is not always clock auction (it is rather an outcry auction).
- ! However, in an IPV environment, 'drops' of the others don't change your own valuation, and it is a weakly undominated strategy to set the limit at your true value.
- Note: In a sub-format (called Japanese auction), you actually cannot re-enter.
- The player with the highest limit wins the object and pays the *second* highest limit price.
- *limit* in English auction = *bid* in SPA

Strategic Equivalence of Auction Formats Under IPV

Proxy bidding in English auction

- e-Bay auctions (and any other online auction, e.g., Sotheby) are open and ascending (English), but bidders are entering *proxy bids* (limits) instead of incrementally increasing the currently announced price.
- Historically, absentee bidders were allowed to submit their limits (*absentee bids*) already in 19th century in stamp auctions; auctioneers were then bidding in English auction on their behalf.
- In IPV environment, imposing proxy bidding instead of present bidding has no effect.
- In IPV environment, choosing proxy bidding instead of present bidding has no effect.

Strategic Equivalence of Auction Formats Under IPV

Open vs. sealed-bid auctions

- The advantage of sealed-bid auctions is that bidders don't have to participate at a specific point of time.
- In an IPV environment, given strategic equivalence, an open (English or Dutch) auction is unnecessary.
- If organized, allowing for proxy bidding is without consequences. The proxy bidders are not disadvantaged.
- ! Under interdependent/correlated values, the strategic equivalence is preserved between Dutch auction and FPA, but not between English auction and SPA.

Auction Formats: Briefly on Collusion

- Take a ring (cartel) within a *single* auction. What is a consequence of defection?
- Except for English auction, a successful defection ends auction (the cartel is dissolved).
- In English auction, where auction doesn't end, the cartel may have reputation motive to punish defecting members to preserve discipline of the ring in the next stages.
- Dutch: Unless a proxy bidding tool is used, an increase in the limit that didn't win the auction is undetectable.

... collusion is hardest to support when secret price concessions are possible, and easiest to support when all price offers must be made publicly. (Milgrom, 1987)

**Grau, teurer Freund, ist alle Theorie, und grün des Lebens goldner
Baum. (Faust)**

Gray is all theory, green is the tree of life.

Main auction formats disregard ...

- credibility issue (the buyer's *default* on bid; the auctioneer's *refusal* to sell)
- reserve prices (minimal bids)
- richer mechanisms, e.g., a possibility for the seller to give a direct offer prior to the auction, or a Buy-It-Now option available jointly with the auction
- cashless bidders (liquidity constraints); when the seller's payment is deferred (through a sequence of uncertain future payments), the seller is bearing a risk and there might be *adverse selection* problem (low-value bidders facing higher default rates bid more aggressively)

Main information environments disregard ...

- costly cognition (especially with constructing complex bidding strategies)
- information acquisition decisions (own value, the opponents' signals)